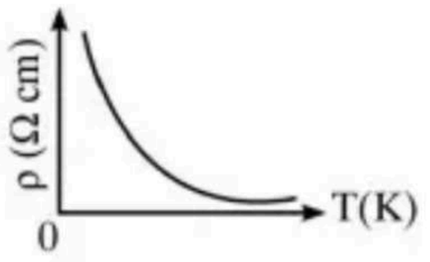
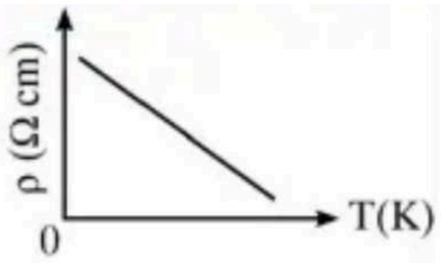
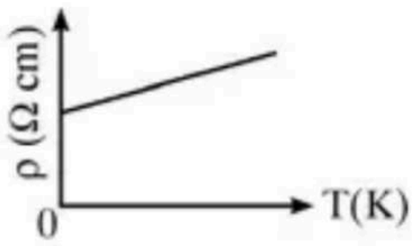
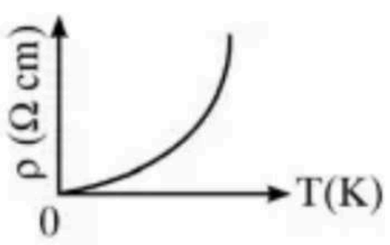
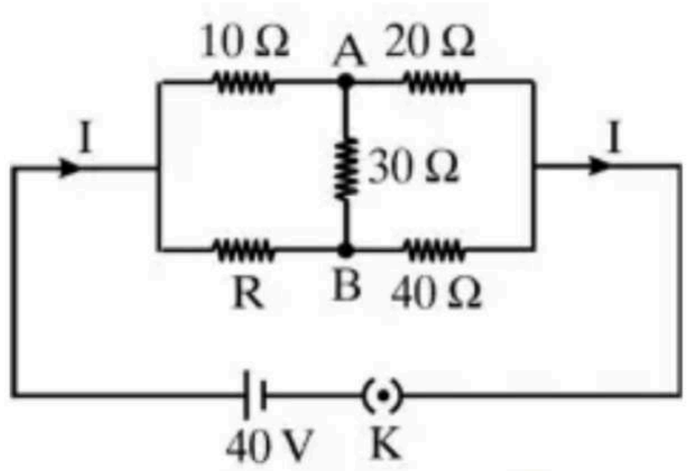
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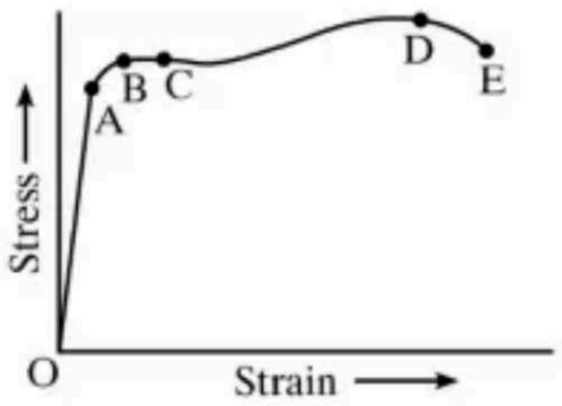








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Stress
Strain
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A
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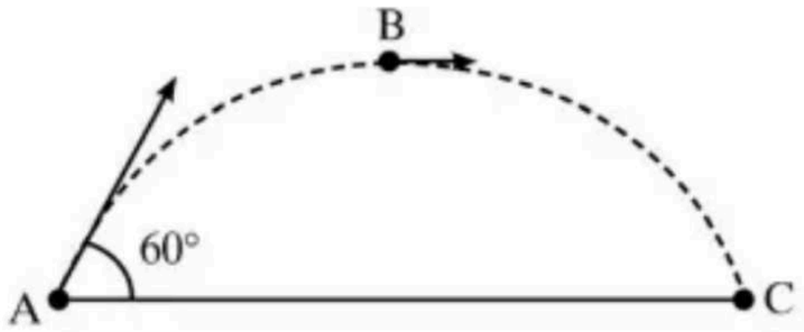
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26. Match List I with List II.

	List I		List II
A.	Excess pressure inside a liquid drop	I.	$P_0+\frac{2T}{R}$
B.	Excess pressure inside a soap bubble	II.	$\frac{2T}{R}$
C.	Pressure inside an air bubble in water at depth 'h'	III.	$\frac{4T}{R}$
D.	Pressure inside a liquid drop	IV.	$P_0+hdg+\frac{2T}{R}$

Choose the correct answer from the options given below.

(1) A-II B-III C-I D-IV

(2) A-II B-III C-IV D-I

(3) A-I B-III C-IV D-II

(4) A-III B-II C-IV D-I

27. Given below are two statements.

Statement I : The idealised reversible process is a quasi-static process with no dissipative factors such as friction, viscosity, etc.

Statement II: The spontaneous process of nature is irreversible.

In light of the above statements, choose the correct answer from the options given below.

(1) Statement I is correct, but statement II is incorrect.

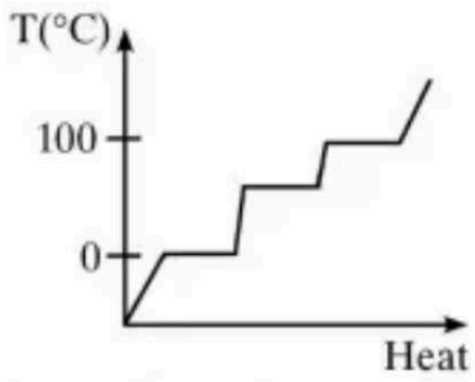
(2) Statement I is incorrect, but statement II is correct.

(3) Both statement I and statement II are correct.

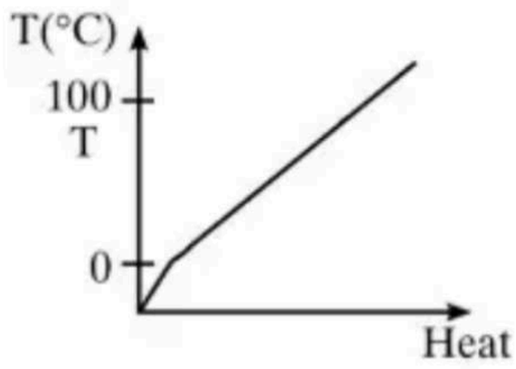
(4) Both statement I and statement II are incorrect.

28. Which of the following best represents the temperature versus heat supplied graph for water, in the range of $-20°C$ to $120°C$

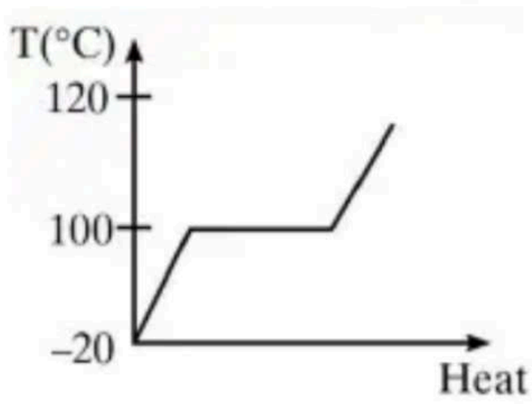
(1)



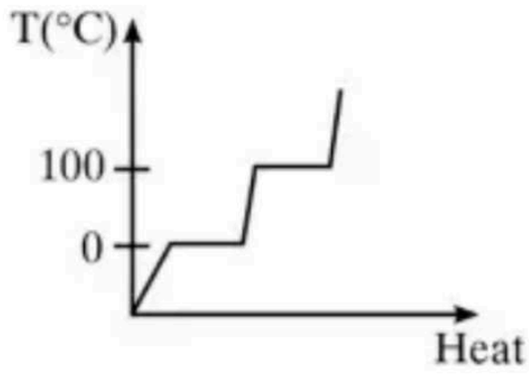
(2)



(3)



(4)



29. The kinetic energy of translation of the molecules in 50 g of CO_2 gas at 17°C is

(1) 3986.3 J

(2) 4102.8 J

(3) 4205.5 J

(4) 3582.7 J

30. A concave mirror of focal length f in the air is dipped in a liquid of refractive index μ. Its focal length in the liquid will be

(1) $\frac{f}{\mu}$ (2) $\frac{f}{(\mu-1)}$

(3) μf (4) f

31. Each side of a thin convex lens of glass (refractive index μ) has radius of curvature R. One side is polished for complete reflection. At what distance from the lens, should an object be placed on the optic axis so that the image gets formed on the object itself?

(1) R/μ (2) $R/(2\mu-3)$

(3) μR (4) $R/(2\mu-1)$

32. A double slit interference experiment performed with a light of wavelength 600 nm forms an interference fringe pattern on a screen with 10th bright fringe having its centre at a distance of 10 mm from the central maximum. The distance of the centre of the same 10th bright fringe from the central maximum when the source of light is replaced by another source of wavelength 660 nm would be _____mm

(1) 11 mm

(2) 16 mm

(3) 9 mm

(4) 6 mm

33. An electron is made to enter symmetrically between two parallel metal plates, each of 10 cm length, having equal and opposite charges. The electron emerges out of the plates with a horizontal component of velocity 10^6 m/s. If the magnitude of the electric field between the plates is 9.1 V/cm, then the vertical component of the velocity of electron is

(mass of electron $=9.1\times10^{-31}$ kg and charge of electron $=1.6\times10^{-19}$C)

(1) 1×10^6 m/s

(2) Zero

(3) 16×10^6 m/s

(4) 16×10^4 m/s

34. A time varying potential difference is applied between the plates of a parallel plate capacitor of capacitance 2.5 μF. The dielectric constant of the medium between the capacitor plates is 1. If it produces an instantaneous displacement current of 0.25 mA in the intervening space between the capacitor plates, the magnitude of the rate of change of the potential difference will be____ Vs^{-1}

(1) 200 (2) 6

(3) 400 (4) 100

35. A coil of area A and N turns is rotating with angular velocity ω in a uniform magnetic field $\vec{B}$ about an axis perpendicular to $\vec{B}$. The magnetic flux φ and the induced emf ε across it, at an instant when $\vec{B}$ is parallel to the plane of the coil, are

(1) $\varphi=AB,\ \varepsilon=0$

(2) $\varphi=0,\ \varepsilon=NAB\omega$

(3) $\varphi=0,\ \varepsilon=0$

(4) $\varphi=AB,\varepsilon=NAB\omega$

36. A complex current wave is given by $i=5+5\sin(100\omega t)$ A. Its average value and RMS value over one time period, respectively, are

(1) 10 A, 10 A

(2) 5 A, $\sqrt{37.5}$ A

(3) $\sqrt{50}$ A, 0

(4) 0, 0

37. Of the various series of the hydrogen spectrum, the one that lies wholly in the ultraviolet region is

(1) Lyman series

(2) Balmer series

(3) Paschen series

(4) Brackett series

38. The binding energies per nucleon for a deuteron and an $\propto$-particle are 1.1 MeV and 7.1 MeV respective. The energy released in the reaction $_1H^2 + _1H^2 \rightarrow _2He^4$ is Q. Find the value of $\frac{Q}{3}$ in MeV.

(1) 12

(2) 8

(3) 15

(4) 24

39. Curie temperature is the temperature above which

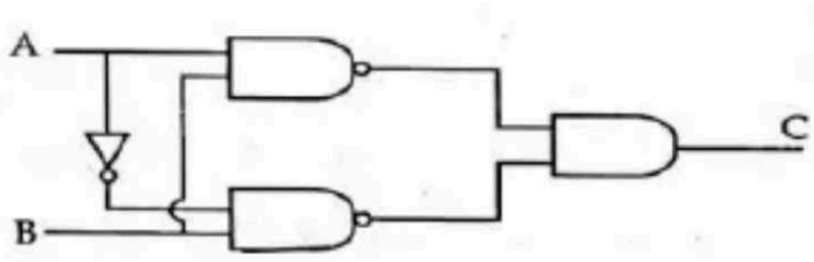
(1) a ferromagnetic material becomes paramagnetic

(2) a paramagnetic material becomes diamagnetic

(3) a ferromagnetic material becomes diamagnetic

(4) a paramagnetic material becomes ferromagnetic

40. The correct truth table for the given logic circuit is



(1)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

(2)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

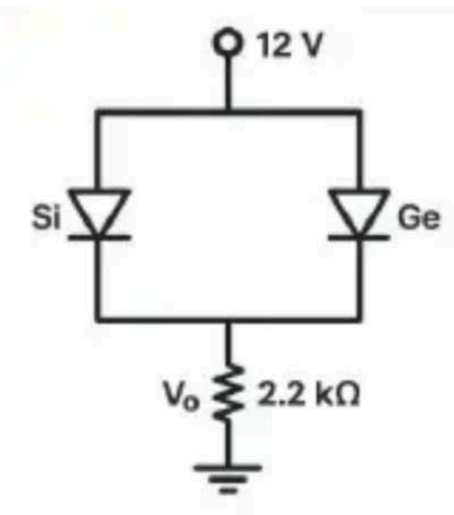
(3)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

(4)

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

41. In the circuit shown in the figure below, voltage V_0 is



(1) 11.7 volt

(2) 11.3 volt

(3) 3.0 volt

(4) 11 volt

42. If the two half-power frequencies for a series L-C-R circuit is 40 Hz and 90 Hz, the resonance frequency is

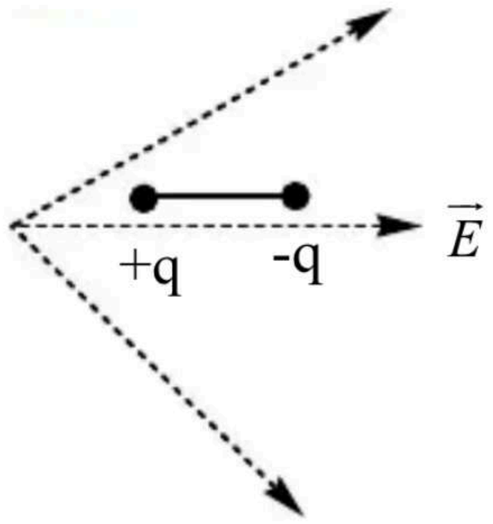
(1) 55.38 Hz

(2) 30 Hz

(3) 50 Hz

(4) 60 Hz

43. A dipole is placed in an electric field as shown in the figure below. Find the direction in which it moves?



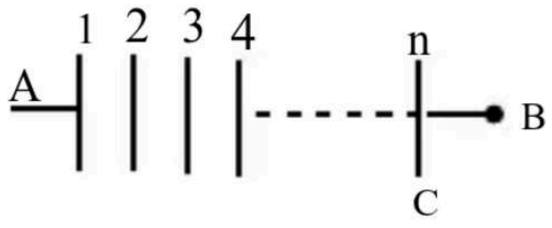
(1) Towards the left, as its potential energy will increase.

(2) Towards the right, as its potential energy will decrease.

(3) Towards the left, as its potential energy will decrease.

(4) Towards the right, as its potential energy will increase.

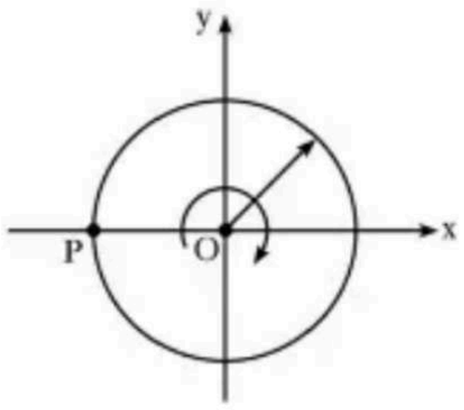
44. Find the equivalent capacitance between points A and B if the capacitance between any two plates is C.



(1) $\frac{C}{2n-1}$ (2) $\frac{C}{2n+1}$

(3) $\frac{C}{n+1}$ (4) $\frac{C}{n-1}$

45. A ring rotates about the z-axis as shown in the figure. The plane of rotation is xy. At a certain instant, the acceleration of a particle P (shown in the figure) on the ring is $(6\hat{i}-8\hat{j})\ m/s^2$. Find the magnitudes of the angular acceleration of the ring and its angular velocity at that instant. (The radius of the ring is 2 m.)



(1) $4\ rad/s^2, \sqrt{3}\ rad/s$

(2) $2\ rad/s^2, 3\ rad/s$

(3) $3\ rad/s^2, 2\ rad/s$

(4) $5\ rad/s^2, \sqrt{5}\ rad/s$

46. The correct statements among the following are

A. Order of melting points of hydrogen halides is HCl > HBr > HF > HI.

B. Order of stability of oxides of halogens is Cl > Br > I.

C. Fluorine is the most electronegative element and cannot exhibit any positive oxidation state.

D. Seaweeds and chile saltpetre are sources of iodine.

E. Order of oxidising powers of halogens is $F_2 > Cl_2 > Br_2 > I_2$

(1) C, D, E only

(2) B, C, D, E only

(3) A, C, D only

(4) B, C, E only

47. Select the correct statements.

A. The dissociation constant of water is less than H_2S at 298K.

B. All hexafluorides of 16th group elements are gaseous in nature.

C. Covalency of sulphur in rhombic sulphur is '6'.

D. In a compound formation, where the sulphur atom is in the 2nd excited state, the possible hybridisation sulphur in that compound will be sp^2 or sp^3 or sp^3d^2.

(1) Only A, B, D (2) Only A, C

(3) Only A, B, C (4) Only C, D

48. Given below are two statements.

Statement I : $T\Delta S_{system}$ is the energy that is not available to do useful work.

Statement II: Absolute entropy is determined using the third law of Thermodynamics, which sets the entropy of a perfect crystal to zero at 0 K.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

49. For the process $A(g)+B(g)\longrightarrow C(g)$, $\Delta H^{\circ}=-30.3\,KJ/mole$. If S°_{A}, S°_{B}, and S°_{C} are $144\,JK^{-1}\,mole^{-1}$, $81\,JK^{-1}\,mole^{-1}$, and $125\,JK^{-1}\,mole^{-1}$, respectively, is the process spontaneous?

(1) No, $\Delta G^{\circ}=+0.5\ KJ/mole$

(2) Yes, $\Delta G^{\circ}=-60.1\ KJ/mole$

(3) Yes, $\Delta G^{\circ}=-0.5\ KJ/mole$

(4) No, $\Delta G^{\circ}=+60.1\ KJ/mole$

50. Given below are two statements.

Statement I: BF_6^{3-} cannot exist.

Statement II: $E^{\circ}_{Tl^{3+}/Tl}$ is positive.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

51. The number of atoms present in 98 grams of H_2SO_4 is (x) and the number of atoms present in 98 grams of H_3PO_4 is (y). The correct relation between them is

(1) $x=y$

(2) $y>x$

(3) $x>y$

(4) cannot be predicted

52. The decomposition of a certain mass of $CaCO_3$ gave 11.2 dm^3 of CO_2 gas at STP. The mass of KOH required to completely neutralise the gas is

(1) 28 g (2) 42 g

(3) 56 g (4) 20 g

53. Given the standard electrode potentials:

$E^{\circ}_{Zn^{2+}/Zn}=-0.76\,V$; $E^{\circ}_{Cu^{2+}/Cu}=+0.34\,V$

Calculate the standard Gibbs free energy change (ΔG°) for the reaction:

$Zn(s)+Cu^{2+}(aq)\rightarrow Zn^{2+}(aq)+Cu(s)$

(1) – 106.2 kJ

(2) – 180.9 kJ

(3) – 342.6 kJ

(4) – 212.4 kJ

54. Given below are two statements.

Statement I : $\Lambda^{\circ}_{eq}(BaCl_2)=\lambda^{\circ}_{eq}(Ba^{+2})+\lambda^{\circ}_{eq}(Cl^{-})$

Statement II: $\Lambda^{\circ}_{m}(H_2O)=\Lambda^{\circ}_{m}(HCl)+\Lambda^{\circ}_{m}(NaOH)-\Lambda^{\circ}_{m}(NaCl)$

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

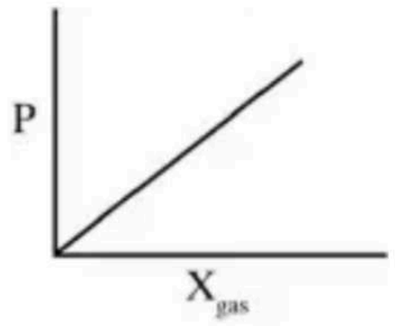
(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

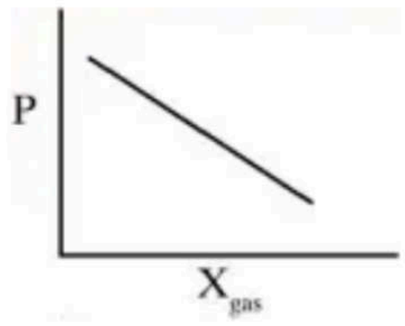
(4) Statement I is incorrect, but statement II is correct.

55. Which one of the following represents Henry's law?

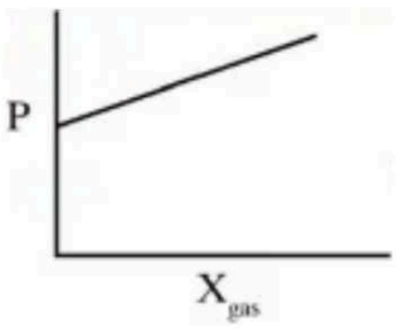
(1)



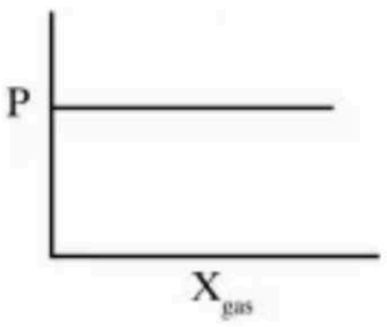
(2)



(3)



(4)



56. Y g of a non-volatile organic substance of molecular mass M is dissolved in 250 g of benzene. The molal elevation constant of benzene is K_b. Elevation in its boiling point is given by

(1) $\frac{M}{K_b Y}$ (2) $\frac{4K_b Y}{M}$

(3) $\frac{K_b Y}{4M}$ (4) $\frac{K_b Y}{M}$

57. The equilibrium constant for the reaction $N_{2(s)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ is 4×10^{-4} at 200 K. In the presence of a catalyst, the equilibrium is attained 10 times faster. Therefore, the equilibrium constant in the presence of a catalyst at 200 K is

(1) 40×10^{4} (2) 4×10^{-4}

(3) 4×10^{3} (4) 2×10^{-3}

58. How many of the following molecules violate(s) the octet rule?

NH_3, $AlCl_3$, $BeCl_2$, CCl_4, PCl_5

(1) 4 (2) 1

(3) 3 (4) 5

59. Match List I with List II.

	List I		**List II**
A.	PCl_5	I.	Square pyramidal
B.	SF_6	II.	Trigonal planar
C.	BrF_5	III.	Octahedral
D.	BF_3	IV.	Trigonal bipyramidal

Choose the correct answer from the options given below.

(1) A-IV B-III C-II D-I

(2) A-IV B-III C-I D-II

(3) A-II B-III C-IV D-I

(4) A-III B-I C-IV D-II

60. The energies E_1 and E_2 of two radiations are 25 eV and 50 eV, respectively. The relation between their wavelengths λ_1 and λ_2 will be

(1) $\lambda_1 = \lambda_2$ (2) $\lambda_1 = 2\lambda_2$

(3) $\lambda_1 = 4\lambda_2$ (4) $\lambda_1 = \frac{1}{3}\lambda_2$

61. For the reaction $2N_2O_5 \longrightarrow 4NO_2 + O_2$, rate and rate constant are 1.02×10^{-4} mole. $lit^{-1}.sec^{-1}$ and 3.4×10^{-5} sec^{-1} respectively. Then, the concentration of N_2O_5 at that time will be _____ M.

(1) 1.732 (2) 3

(3) 1.02×10^{-4} (4) 3.4×10^{5}

62. The solubility of calcium phosphate (molecular mass = M) in water is W g per 100 mL at 25°C. Its solubility product at 25°C will be

(1) $1.08\times10^{3}\,(W/M)^{5}$

(2) $1.08\times10^{5}\,(W/M)^{2}$

(3) $1.08\times10^{7}\,(W/M)^{5}$

(4) $1.08\times10^{2}\,(W/M)^{4}$

63. Which of the following constitute the basic radicals of the fourth group in qualitative analysis?

(1) Pb^{2+}, Hg^{2+} (2) Zn^{2+}, Mn^{2+}

(3) Al^{3+}, Fe^{3+} (4) Ca^{2+}, Ba^{2+}

64. The mechanism of a hypothetical reaction, $X_2 + Y_2 \rightarrow 2XY$, is given below.

A. $X_2 \rightleftharpoons X + X\,(\text{fast})$

B. $X + Y_2 \rightarrow XY + Y\,(\text{slow})$

C. $X + Y \rightarrow XY\,(\text{fast})$

The overall order of the reaction will be

(1) 2 (2) 0

(3) 1.5 (4) 1

65. Which of the following statements are correct?

A. For a given principal quantum number, the total number of 'm' values are n^2.

B. For a given principal quantum number, the total number of 'ℓ' values are $(n-1)$.

C. For a given azimuthal quantum number, the total number of electrons are $(4\ell+2)$.

D. For a given magnetic quantum number, the total number of electrons are $2n^2$.

(1) Only B and C

(2) Only A and B

(3) Only A

(4) Only A and C

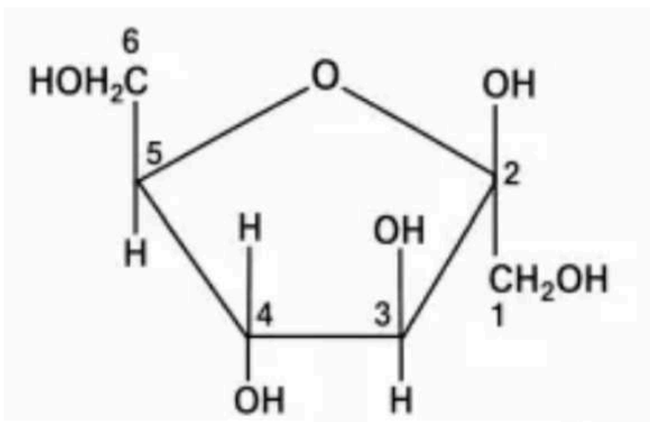
66. Lyophobic colloid among the following is

(1) Starch

(2) Gelatin

(3) $Fe(OH)_3$

(4) Gum

67. 

The given cyclic structure corresponds to

(1) $\alpha-D-(+)-$Glycopyranose

(2) $\beta-D-(+)-$Glucopyranose

(3) $\beta-D-(-)-$Fructofuranose

(4) $\alpha-D-(-)-$Fructofuranose

68. K_a for HCN is 5×10^{-10}. For maintaining a constant pH of 9.6020, the volume of 5M KCN solution required to be added to 10 ml of 2M HCN solution is

(1) 2 ml

(2) 4 ml

(3) 8 ml

(4) 16 ml

69. Which of the following amine can be prepared by the Hoffmann bromide reaction but not by the Gabriel Phthalimide reaction?

(1) CH_3NH_2

(2) $Me-NH-Me$

(3) Me_3N

(4) $Ph-NH_2$

70. Match List I with List II.

	List I		**List II**
A.	Pnicogen	I.	Ts
B.	Chalcogen	II.	Og
C.	Halogen	III.	Lv
D.	Noble gas	IV.	Mc

Choose the correct answer from the options given below.

(1) A-IV B-I C-II D-III

(2) A-IV B-III C-I D-II

(3) A-III B-I C-IV D-II

(4) A-II B-III C-IV D-I

71. The number of structural isomers possible for the molecular formula $C_5H_{11}Br$ is

(1) seven (2) eight

(3) six (4) nine

72. All the following are examples of electrophilic substitution, except

(1) Reimer – Tiemann reaction

(2) Sandmeyer's reaction

(3) Coupling reaction of phenol with benzene diazonium chloride

(4) Friedel – Craft's alkylation

73. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Unlike phenols, carboxylic acids react with weaker bases such as carbonates and hydrogen carbonates to evolve carbon dioxide.

Reason (R) : Carbonates and Bicarbonates are weak bases and are not strong enough to pull the proton of phenol, which is a weaker acid than carboxylic acids.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

74. In which of the following reactions is a vicinal diol obtained?

(1) 2-butene $\xrightarrow[273K]{alk.KMnO_4}$

(2) 2-butene $\xrightarrow[Zn/H_2O]{O_3}$

(3) 2-butene $\xrightarrow{KMnO_4/H^+}$

(4) 2-butene $\xrightarrow{Br_2/H_2O}$

75. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Reactivity of the following bromides towards S_N1 is

$(CH_3)_3CBr$

$>CH_3CH(Br)CH_2CH_3$

$>CH_3CH_2CH_2CH_2Br$

Reason (R) : Order of stability of cations thus formed from these bromides is

$(CH_3)_3\overset{+}{C} > CH_3\overset{+}{C}HCH_2CH_3$

$>CH_3CH_2CH_2\overset{+}{C}H_2$

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

76. Match List I with List II.

	List I (Complex)		**List II (Colour of complex)**
A.	$[Ni(H_2O)_4(en)]^{2+}$	I.	Green
B.	$[Ni(H_2O)_2(en)_2]^{2+}$	II.	Violet
C.	$[Ni(en)_3]^{2+}$	III.	Blue/Purple
D.	$[Ni(H_2O)_6]^{3+}$	IV.	Pale blue

Choose the correct answer from the options given below.

(1)	A-IV	B-III	C-II	D-I
(2)	A-IV	B-III	C-I	D-II
(3)	A-IV	B-I	C-II	D-III
(4)	A-IV	B-II	C-I	D-III

77. In the 13th group, the element with least melting point is

(1) B (2) Al

(3) Ga (4) In

78. $CH_3MgBr \xrightarrow[H_2O]{HCHO} A \xrightarrow{\text{thionyl chloride}} B \xrightarrow[b)\,H_3O^+]{a)\,alc.KOH} \underset{(final)}{C}$.

The isomerism exhibited by 'C' is

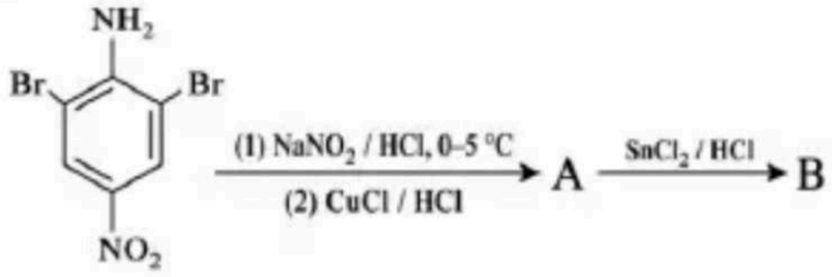
(1) chain isomerism

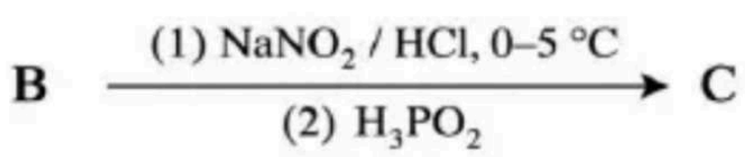
(2) positional isomerism

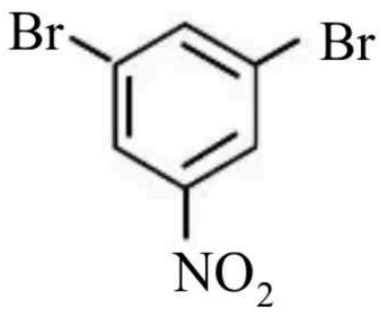
(3) functional isomerism

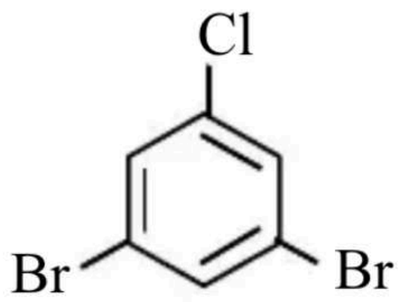
(4) metamerism

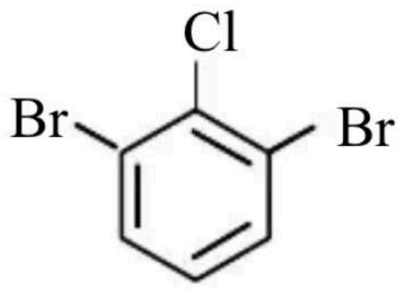
79. The product (C) obtained in the following sequence of reactions is

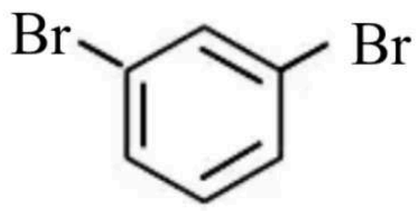




(1) 

(2) 

(3) 

(4) 

80. Identify the correct position and nature of the element with atomic number 32 in the modern periodic table.

(1) 3rd period, 14th group and metalloid

(2) 4th period, 14th group and metalloid

(3) 4th period, 13th group and metal

(4) 3rd period, 13th group and metal

81. Given below are two statements.

Statement I : Benzoic acid does not undergo Friedel–Craft's reaction.

Statement II: Benzoic acid on nitration forms m–nitro benzoic acid.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

82. Which of the following statements are correct?

A. Benzene can involve in electrophilic substitution reactions.

B. Resonance energy of benzene is 36 $k.cal.mol^{-1}$.

C. Chlorobenzene is more reactive than benzene towards the electrophilic substitution reaction.

D. In arenium ion, delocalisation of electrons stops at sp^3 hybridised carbon.

(1) A, B, and C

(2) A, D, and C

(3) A, B, and D

(4) A, B, C, and D

83. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : $[Ti(H_2O)_6]^{3+}$ is coloured.

Reason (R) : d-d transition is possible in $[Ti(H_2O)_6]^{3+}$ due to an unpaired d electron.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

84. The correct statement(s) among the following is/are

A. The actinoids are radioactive elements, and the earlier members have relatively longer half-lives than the latter ones.

B. Electronic configuration of the cerium atom (Z=58) is $[Xe]6s^2\,5d^o\,4f^2$.

C. Ionic radius of $Eu^{3+} > Ce^{3+}$

(1) Only A and B

(2) Only B and C

(3) Only A and C

(4) Only A

85. Choose the INCORRECT statement.

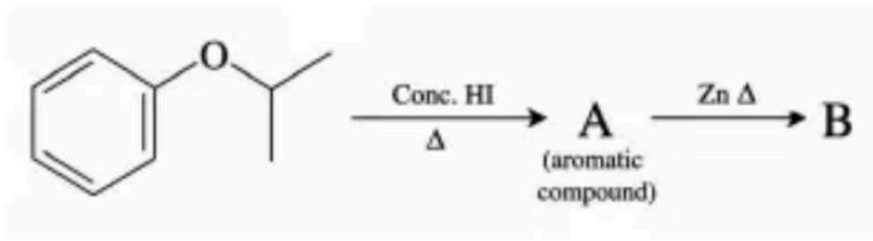
(1) In denaturation of alcohol, copper sulphate and pyridine are added.

(2) In the conversion of glucose to ethyl alcohol, the enzyme zymase is involved.

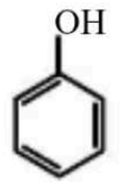
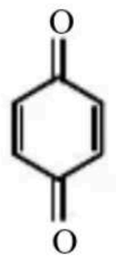
(3) In the preparation of methanol from a mixture of carbon monoxide and hydrogen, the catalyst $ZnO-Cr_2O_3$ is used.

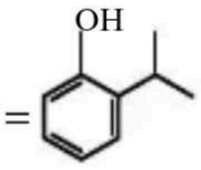
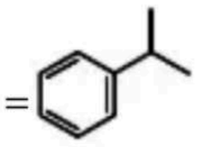
(4) Butan –1–ol < butan–2–ol < Propan – 1–ol is the increasing order of boiling point.

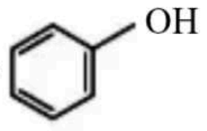
86. Ether is heated with conc. HI to give a hydroxy compound A which is further heated with Zn dust to give compound B. Identify A and B.



(1) A =  , B = 

(2) A =  , B = 

(3) A =  , B = 

(4) A =  , B = 

87. Identify the following statements as True (T) or False (F).

A. P-Amino azobenzene is an orange dye.

B. The yield in the Gattermann reaction is found to be better than the Sandmeyer reaction.

C. N-Ethyl benzene sulphonamide is soluble in dil. NaOH.

D. Benzene diazonium fluoroborate is stable at room temperature and insoluble in water.

(1) T F T T

(2) T F T F

(3) F F T T

(4) F T F T

88. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : The staggered form of ethane is more stable than the eclipsed form.

Reason (R) : Torsional strain in the staggered conformation of ethane is less than in the eclipsed form.

In light of the above statements, choose the correct answer from the options given below.

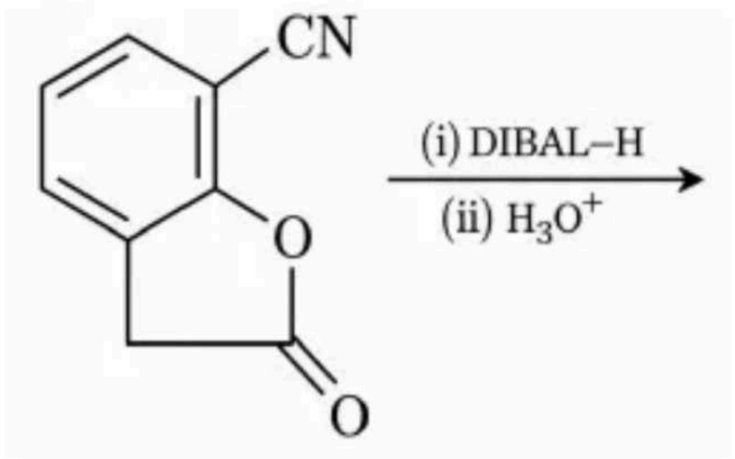
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

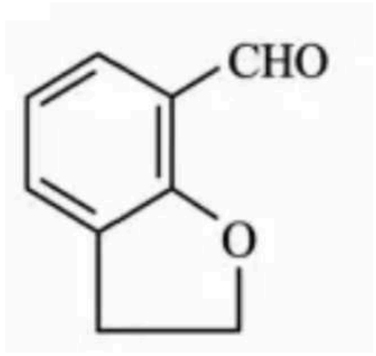
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

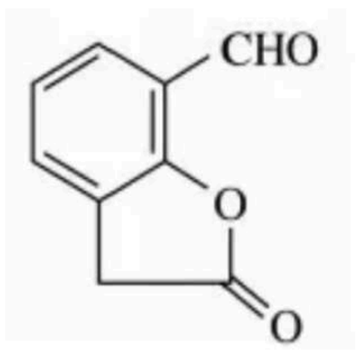
(3) (A) is true, but (R) is false.

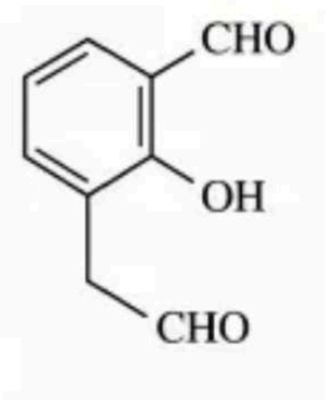
(4) (A) is false, but (R) is true.

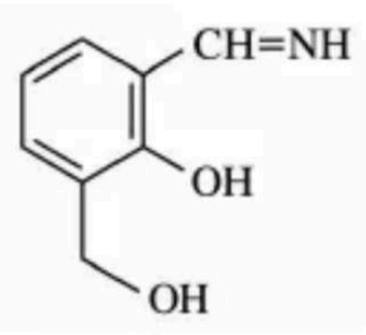
89. The major product of the following reaction is



(1) 

(2) 

(3) 

(4) 

90. All the following will show -I effect, except

(1) $-O^-$ (2) $-COOH$

(3) $-\overset{+}{S}R_2$ (4) $-OCH_3$

91. The separated bands of DNA are cut out from agarose gel and extracted from gel piece in a process known as

(1) elution (2) spooling

(3) purification (4) staining

92. Which of the following statements are correct regarding the menstrual cycle?

A. The first menstruation begins at puberty and is called menarche.

B. The ovulatory phase is followed by the proliferative phase.

C. Menstruation occurs if the released ovum is not fertilised.

D. Cyclic menstruation extends between menarche and menopause.

E. During pregnancy, all events of the menstrual cycle stop, and there is no menstruation.

Choose the most appropriate answer from the options given below.

(1) Only A and B

(2) Only A, B, and C

(3) Only A, C, D, and E

(4) A, B, C, and D

93. Given below are two statements.

Statement I : *Agrobacterium tumefaciens*, a pathogen of several dicots, is able to deliver a piece of DNA known as 'T-DNA' to transform a normal plant cell into a tumour.

Statement II: Organic farming is a holistic approach that seeks to develop an understanding of the web of interaction between the myriad of organisms that constitute the field's fauna and flora.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

94. Given below are two statements.

Statement I : Amniocentesis is a procedure that can be used to test for the presence of certain genetic disorders such as sickle-cell anaemia and haemophilia to determine the survivability of the foetus.

Statement II : In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the foetal cells and dissolved substances.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

95. Which of the following statements are correct?

A. In green algae, sexual reproduction may be isogamous, anisogamous, or oogamous.

B. In brown algae, fusion of gametes occurs in water, only not inside the oogonium.

C. In liverworts, sporophyte remains attached to gametophyte and depends on it for nutrition.

D. In pteridophytes, gametophyte (prothallus) bears sporangia on sporophylls.

E. In pteridophytes, zygote develops into embryo inside archegonium.

(1) A, B, C, and E only

(2) A, C, and E only

(3) B, C, D, and E only

(4) A, B, D, and E only

96. Study the following:

$$GPP - R = NPP$$

In the above equation, NPP represents

(1) the biomass utilised by plants in respiration.

(2) biomass available to heterotrophs (herbivores and decomposers) for consumption

(3) the rate of formation of new organic matter by consumers

(4) the rate of production of organic matter during photosynthesis

97. Match List I with List II.

	List I		List II
A.	Flower colour in garden pea	I.	Codominance
B.	Flower colour in *Antirrhinum*	II.	Complete dominance
C.	Human blood group AB	III.	Incomplete dominance
D.	Skin colour in human	IV.	Polygenic Inheritance

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-I B-III C-IV D-II

(3) A-II B-III C-I D-IV

(4) A-IV B-III C-II D-I

98. Functional Residual Capacity of lungs =

(1) IRV + ERV + TV + RV

(2) IRV + ERV + TV - RV

(3) ERV + RV

(4) IRV + ERV

99. Read the following statements.

A. Photosynthesis is generally a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates.

B. PS II itself is physically located on the inner side of the thylakoid membrane.

C. The NADP reductase enzyme is located on the stroma side of the thylakoid membrane.

D. Complete oxidation of glucose, which produces CO_2 and H_2O as the end products, yields energy, most of which is given out as heat.

Select the correct statements.

(1) A, B, C, and D

(2) Only A, B, and C

(3) Only A and B

(4) Only B and C

100. Itching, fluid discharge, slight pain, swellings, etc., in the genital region, are the minor and early symptoms of

(1) most of the STIs

(2) all helminth infections

(3) most congenital genetic disorders

(4) all fungal infections

101. Given below are two statements. One islabelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Plant growth and further development are intimately linked to the water status of the plant.

Reason (R) : Turgidity of cells helps in extension growth.

In light of the above statements, choose the correct answer from the options given below.

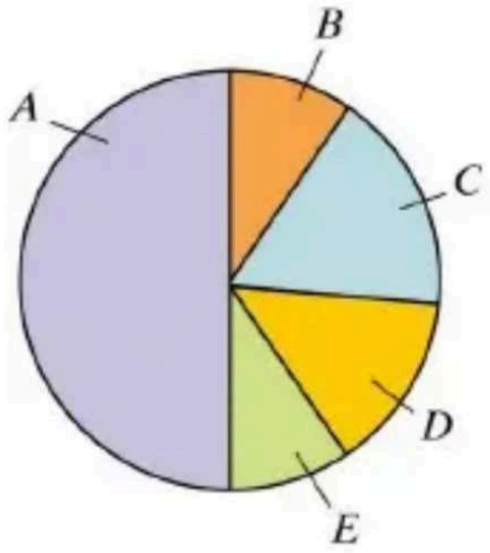
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

102. Observe the diagram below and select the correct option.



(1) *B* – Aves

(2) *D* – Reptiles

(3) *A* – Mammals

(4) *C* – Amphibians

103. Which of the following is under the direct neural regulation of the hypothalamus?

(1) Adrenal cortex

(2) Adenohypophysis

(3) Pars intermedia

(4) Posterior pituitary

104. Identify the false feature for the given organism.



(1) Rhodophycean algae

(2) Oogamous sexual reproduction

(3) Flagellated structures absent in life cycle

(4) Reserve food material is mannitol

105. The sexual stages (gametocytes) of *Plasmodium* develop in

(1) human red blood cells

(2) haemolymph of the mosquito

(3) human liver cells

(4) mouth parts of the mosquito

106. In Kranz anatomy, the bundle sheath cells have

(1) thin walls, many intercellular spaces, and no chloroplasts

(2) thick walls, no intercellular spaces, and a large number of chloroplasts

(3) thin walls, no intercellular spaces, and several chloroplasts

(4) thick walls, many intercellular spaces, and few chloroplasts

107. Given below are two statements.

Statement I : Mitosis cannot occur without DNA replication (in the S phase).

Statement II : Exceptionally, there can be DNA replication without cell division (polyploidy).

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

108. Match List I with List II.

	List I		List II
A.	SAN	I.	Ventricular musculature
B.	Purkinje fibres	II.	Right upper corner of the right atrium
C.	AVN	III.	Emerges on the top of the interventricular septum
D.	AV bundle	IV.	Lower left corner of the right atrium

Choose the correct answer from the options given below.

(1) A-II B-I C-III D-IV

(2) A-I B-II C-IV D-III

(3) A-I B-II C-III D-IV

(4) A-II B-I C-IV D-III

109. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Superclass Pisces and Superclass Tetrapoda are included under Gnathostomata.

Reason (R) : The members of Pisces and the members of Tetrapoda possess a mouth provided with jaws.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

110. Given below are two statements.

Statement I : Ligaments are dense, regular connective tissue.

Statement II: Cartilage is one of the specialised connective tissues.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is correct, but statement II is correct.

111. In monoglycerides, glycerol is attached to a fatty acid molecule by a/an

(1) ester bond

(2) peptide bond

(3) glycosidic bond

(4) phosphodiester bond

112. Match List I with List II.

	List I		List II
A.	Indole compounds	I.	Seed dormancy promotion
B.	Adenine derivatives	II.	Overcome apical dominance
C.	Carotenoids derivatives	III.	Delay the senescence of fruits to extend the market period
D.	Terpene derivatives	IV.	Preparing a weed-free lawn

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-IV B-III C-II D-I

(3) A-IV B-II C-III D-I

(4) A-IV B-II C-I D-III

113. An individual has overall masculine development. However, gynaecomastia is also expressed. Such an individual exhibits

(1) Turner's Syndrome

(2) Klinefelter's Syndrome

(3) Edward's Syndrome

(4) Down's Syndrome

114. Arrange the following events of development in chronological order.

A. Plasmatic growth

B. Senescence

C. Elongation

D. Maturation

E. Differentiation

(1) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

(2) $A \rightarrow C \rightarrow E \rightarrow D \rightarrow B$

(3) $A \rightarrow C \rightarrow D \rightarrow E \rightarrow B$

(4) $A \rightarrow D \rightarrow C \rightarrow E \rightarrow B$

115. Which of the following statements is correct in relation to the frog's heart?

(1) It is neurogenic.

(2) It has two ventricles and one atrium.

(3) A triangular structure called sinus venosus joins the right atrium.

(4) The ventricle opens into a sac-like conus arteriosus located on the dorsal side of the heart.

116. Potato and brinjal do NOT belong to the same

(1) species

(2) family

(3) order

(4) genus

117. Given below are two statements.

Statement I : Cancer-causing viruses called oncogenic viruses have genes called viral oncogenes.

Statement II : Antibodies against cancer-specific antigens are also used for the detection of certain cancers.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

118. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Continued self-pollination results in inbreeding depression.

Reason (R) : Majority of flowering plants produce unisexual flowers, and pollen grains are likely to come in contact with the stigma of the same flowers.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

119. Given below are two statements.

Statement I : Vasa efferentia leave the testis and open into the epididymis located along the posterior surface of each testis.

Statement II: The shape of the uterus is like an inverted pear, and it is supported by ligaments attached to the pelvic wall.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

120. The best stage to study the morphology of the chromosome is

(1) interphase

(2) telophase

(3) metaphase

(4) prophase

121. Study the following statements about the members of the kingdom Animalia.

A. Echinoderms exhibit radial or bilateral symmetry depending on the stage.

B. Poriferans and cnidarians exhibit a cellular level of organisation only.

C. The members of Platyhelminthes are acoelomates.

D. The members of Annelida, Arthropoda and Mollusca are coelomates.

Select the correct option.

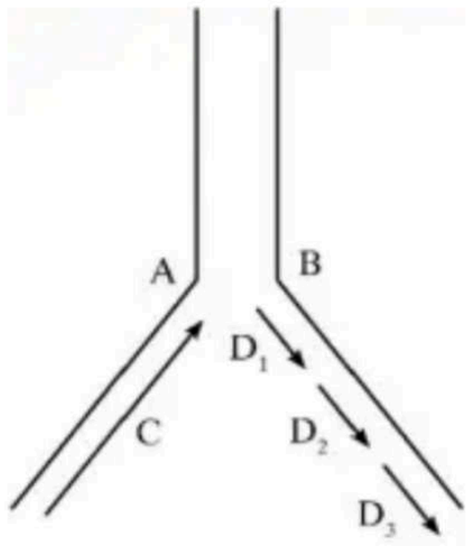
(1) Only A and C are true.

(2) Only B and D are true.

(3) Only A, C, and D are true.

(4) Only B, C, and D are false.

122. Select the INCORRECT statement about the given scheme.



(1) A – template with polarity 5' → 3'

(2) B – template with polarity 5' → 3'

(3) C – leading strand

(4) D – lagging strand

123. Match List I with List II.

	List I (Type of barrier)		**List II (Example)**
A.	Cellular barrier	I.	Mucus coating of urinogenital tract
B.	Physical barrier	II.	Saliva in the mouth
C.	Cytokine barrier	III.	Neutrophils
D.	Physiological barrier	IV.	Interferons

Choose the correct answer from the options given below.

(1) A-III B-I C-II D-IV

(2) A-III B-I C-IV D-II

(3) A-IV B-I C-III D-II

(4) A-III B-II C-IV D-I

124. Match List I with List II.

	List I		List II
A.	Autonomic neural system	I.	Relays impulses from CNS to skeletal muscles
B.	Chemical synapse	II.	Membranes of pre- and post-synaptic neurons are separated by a synaptic cleft.
C.	Somatic neural system	III.	Membranes of pre- and post- synaptic neurons are in very close proximity.
D.	Electrical synapse	IV.	Transmits impulses from the CNS to involuntary organs

Choose the correct answer from the options given below.

(1) A-I B-II C-IV D-III
(2) A-IV B-III C-I D-II
(3) A-I B-II C-III D-IV
(4) A-IV B-II C-I D-III

125. What is the probability of obtaining double homozygous plants with yellow coloured, and wrinkled seeds in F_2 generation in a Mendelian dihybrid cross in a pea plant?

(1) 6.25 % (2) 12.5 %
(3) 25 % (4) 50 %

126. Select the correct group/set of Australian marsupials exhibiting adaptive radiation.

(1) Lemur, Flying squirrel, Bobcat
(2) Wombat, Tasmanian wolf, Sugar glider
(3) Numbat, Bobcat, Spotted cuscus
(4) Tasmanian wolf, Marsupial mole, Lemur

127. Match List I with List II.

	List I		List II
A.	*Equisetum*	I.	Monoecious gametophyte
B.	*Salvinia*	II.	Dimorphic shoots (long, dwarf shoots)
C.	*Sphagnum*	III.	Rhizome
D.	*Ginkgo*	IV.	Aquatic heterosporous pteridophyte
E.	*Pinus*	V.	Needle-like leaves

Choose the correct answer from the options given below.

(1) A-I B-IV C-III D-II E-V
(2) A-II B-IV C-V D-I E-III
(3) A-III B-IV C-I D-II E-V
(4) A-III B-IV C-II D-V E-I

128. Given below are two statements.

Statement I : Mendel's concept of factors as stable and discrete units that controlled the expression of traits and, of the pair of alleles which did not blend with each other, was not accepted by his contemporaries as an explanation for apparently continuous variations seen in nature.

Statement II : Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.
(2) Statement I is correct, but statement II is incorrect.
(3) Statement I is incorrect, but statement II is correct.
(4) Both statement I and statement II are incorrect.

129. Match List I with List II.

	List I		List II
A.	Tubectomy	I.	Intrauterine Device
B.	Lactational amenorrhoea	II.	Barrier method
C.	Vaults	III.	Surgical method
D.	LNG-20	IV.	Natural method

Choose the correct answer from the options given below.

(1) A-IV B-II C-I D-III

(2) A-III B-I C-IV D-II

(3) A-III B-IV C-II D-I

(4) A-II B-III C-I D-IV

130. Repeated activation of the muscles can lead to the accumulation of lactic acid due to

(1) aerobic breakdown of glucose in them

(2) Cori cycle in the liver

(3) anaerobic breakdown of glucose/ glycogen in them

(4) deamination of amino acids in blood

131. Which of the following is a bacteriophage that possesses highly unstable, reactive, and rapidly mutating genetic material?

(1) λ phage

(2) QB phage

(3) TMV

(4) T – even phage

132. Identify the correct chronological sequence of the first appearance of the following plant groups in the geological fossil record from earliest to most recent.

A. Seed ferns

B. *Psilophyton*

C. Dicotyledons

D. Progymnosperms

(1) B → C → A → D

(2) B → D → C → A

(3) B → D → A → C

(4) B → A → D → C

133. DNA present in the matrix of mitochondria is

(1) single linear ssDNA

(2) single circular ssDNA

(3) single circular dsDNA

(4) single linear dsDNA

134. Which of the following is now routinely used to detect HIV in suspected AIDS patients?

(1) Polymerase Chain Reaction (PCR)

(2) Widal test

(3) Serum and urine analysis

(4) Magnetic Resonance Imaging (MRI)

135. Read the following statements.

A. tRNA has an anticodon loop that has three bases (anticodon) complimentary to the codon.

B. In the first phase of translation, amino acids are activated in the presence of ATP and linked to their cognate t-RNA.

C. When the small subunit of ribosome encounters an mRNA, the process of translation begins.

D. A translation unit in mRNA is the sequence of RNA that is flanked by the start codon and the stop codon.

Select the correct statements.

(1) Only A, B, and C

(2) A, B, C and D

(3) Only A, C, and D

(4) Only A and B

136. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : During pregnancy the levels of oestrogens, progestogens, cortisol, prolactin, thyroxine, etc. increase several folds in the maternal blood.

Reason (R) : Increased production of these hormones is essential for supporting the foetal growth, metabolic changes in the mother, and maintenance of pregnancy.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

137. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : DNA is preferred for the storage of genetic information.

Reason (R) : DNA is more stable than RNA.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

138. Among the following, how many statements are INCORRECT?

A. Bioreactors can be thought of as vessels in which raw materials are biologically converted into specific products.

B. A bioreactor provides the optimal conditions necessary to achieve the desired product.

C. A stirred-tank reactor has a flat base to facilitate mixing of the reactor contents.

D. The stirrer facilitates even mixing of nutrients and oxygen availability throughout the bioreactor.

E. The bioreactor has an agitator system.

(1) 1 (2) 2

(3) 3 (4) 4

139. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : A fall in glomerular blood flow / glomerular blood pressure / GFR can activate the JG cells to release renin.

Reason (R) : Atrial Natriuretic Factor can cause vasodilation, thereby decreasing the blood pressure.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

140. Milk produced by Rosie, the first transgenic cow, was nutritionally more balanced for human babies due to the presence of which of the following human proteins?

(1) Alpha-1-antitrypsin

(2) Alpha-interferon

(3) Alpha-lactalbumin

(4) Beta-lactalbumin

141. Select the correct sequence of stages of the Calvin cycle.

(1) Reduction → Carboxylation → Regeneration

(2) Carboxylation → Regeneration → Reduction

(3) Carboxylation → Reduction → Regeneration

(4) Reduction → Regeneration → Carboxylation

142. Given below are two statements.

Statement I : Severo Ochoa enzyme (Polynucleotide phosphorylase) can help in polymerising RNA with a defined sequence in a template independent manner.

Statement II: Complementarity exists between nucleotides and amino acids.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

143. Which of the following statements is false in relation to the salient features of human genome?

(1) The average gene consists of 3000 bases.

(2) Less than 2 percent of the genome codes for proteins.

(3) The human genome contains 3164.7 million bp.

(4) The functions are unknown for over 99.9 percent of the discovered genes.

144. Select the correct statements.

A. The detritus food chain begins with dead organic matter.

B. In an aquatic ecosystem, grazing food chain is the major conduit for energy flow.

C. Each trophic level has a certain mass of living material at a particular time called the standing crop.

D. Organisms at each trophic level depend on those at the higher trophic level for their energy demands.

E. The amount of energy increases from lower to upper trophic levels.

(1) Only B, C, and D

(2) Only C, D, and E

(3) Only D, E, and A

(4) Only A, B, and C

145. Which among the following activities is NOT seen in mesophyll cell in C_4 cycle?

(1) RuBisCO activity

(2) C_4 acid formation

(3) PEP carboxylase activity

(4) Synthesis of aspartic acid

146. Which one of the following statements is INCORRECT?

(1) Colombia, located near the equator, has nearly 105 species of birds.

(2) The largely tropical Amazonian rainforest in South America has the greatest biodiversity on earth.

(3) In general, species diversity decreases as we move away from the equator towards the poles.

(4) With very few exceptions, tropics (latitudinal range of 23.5°N to 23.5°S) harbour more species than temperate or polar areas.

147. Which of the following statements are correct?

A. In maize embryo, scutellum is equivalent to cotyledon.

B. The drupe in mango develops from a multicarpellary inferior ovary.

C. Flowers are perigynous in plum, peach, and rose.

D. In *Argemone*, parietal placentation is present.

(1) Only A and B

(2) Only A, C, and D

(3) Only C and D

(4) Only A, B, C, and D

148. Given below are two statements.

Statement I : Due to continental drift, pouched mammals of Australia survived because of the lack of competition from any other mammal.

Statement II: By the time of 500 mya, invertebrates were formed and active.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

149. Effluent from the primary settling tank is sent into '*A*', and sediment from the settling tank is largely sent into '*B*' in STP. What are *A* and *B*?

	A	*B*
(1)	Aeration tank	Large waterbodies
(2)	Aeration tank	Anaerobic sludge digester
(3)	Large waterbodies	Anaerobic sludge digester
(4)	Anaerobic sludge digester	Large waterbodies

150. Which of the following are the last refuges for a large number of rare and threatened plants in Meghalaya?

(1) Nilgiri Hills

(2) Western Ghats

(3) Jaintia Hills

(4) Aravalli Hills

151. How many PGA molecules are formed in Calvin cycle if two molecules of RUBP are carboxylated?

(1) 2 (2) 4

(3) 18 (4) 12

152. Given below are two statements.

Statement I : When Darwin spoke of the struggle for existence and survival of the fittest in nature, he was convinced that interspecific competition is a potent force in organic evolution.

Statement II: Gause's Competitive Exclusion Principle states that two closely related species competing for the same resources cannot co-exist indefinitely, and the competitively inferior one will eventually be eliminated.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

153. Given below are two statements.

Statement I : '*Ori*' is the sequence that not only initiates transcription but is also responsible for controlling the copy number of linked DNA.

Statement II: In order to force bacteria to take up the plasmid, the bacterial cells must first be made 'competent' to take up DNA by treating them with a specific concentration of divalent cations.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

154. Study the following statements about energy flow and identify the true statements.

A. The energy trapped by the producer is either passed on to a consumer or the organism dies.

B. All animals depend on plants directly.

C. The death of an organism is the beginning of the detritus food chain.

D. The consumers that feed on herbivores are primary consumers.

(1) B and D (2) A and D

(3) B and C (4) A and C

155. Making multiple copies of any template DNA inside a host cell is called

(1) polymerisation

(2) transcription

(3) polymerase chain reaction

(4) cloning

156. Which of the following statements are correct about haemophilia?

A. It is a quantitative problem of synthesising too few globin molecules.

B. It is a sex-linked recessive disease.

C. In the affected individual, phenyl pyruvic acid is accumulated in the brain.

D. It is an autosomal dominant disease.

E. The possibility of a female becoming a haemophilic is extremely rare.

(1) Only C and D

(2) Only B and E

(3) Only A, B, and E

(4) Only A, B, and C

157. Identify the correct combinations of plant structures and their respective ploidy levels.

A. Perisperm – 2n

B. PEN – 3n

C. Nucellus – 2n

D. Fertilised egg – n

E. Pollen mother cell – n

(1) Only A and B

(2) Only B, C, and D

(3) Only A and C

(4) Only A, B, and C

158. Which of the following is a correct match?

(1) *Marchantia* : Monoecious gametophyte

(2) *Selaginella* : Heterosporous pteridophyte

(3) *Porphyra* : Flagellated gametes

(4) *Pinus* : Dioecious sporophyte

159. In which abdominal segment of the female cockroach is a pair of spermatheca present?

(1) 9th segment

(2) 5th segment

(3) 4th segment

(4) 6th segment

160. Given below are two statements.

Statement I : The renal tubule begins with a double-walled cup-like structure called Bowman's capsule.

Statement II: Glomerulus is a tuft of capillaries formed by the afferent arteriole – a fine branch of renal artery.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

161. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Meiosis is a mechanism by which conservation of the specific chromosome number of each species occurs across generations in sexually reproducing organisms.

Reason (R) : Meiosis leads to a reduction of chromosome number by half, which is restored by syngamy.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

162. In cockroaches, a ring of 6–8 blind tubules called hepatic caeca is present at the junction of

(1) foregut and midgut

(2) midgut and hindgut

(3) ileum and colon

(4) colon and rectum

163. Read the following statements.

A. Secondary xylem is the major component in the primary dicot stem.

B. The vascular bundles in dicot and monocot leaves are conjoint.

C. The cortex and pith are clearly differentiated in the grass stem.

D. Mesophyll is the ground tissue in leaves.

E. Phloem fibres are usually present in primary phloem.

How many of the above statements are correct?

(1) 3 (2) 1

(3) 4 (4) 2

164. $\frac{dN}{dt} = rN$

The 'r' in this equation represents

(1) population density at time zero

(2) the base of natural logarithms

(3) intrinsic rate of natural increase

(4) carrying capacity

165. In some plants, the leaves of juvenile plants are different in shape from those of mature plants. This type of heterophylly due to different phases of life is seen in

(1) cotton, coriander, and larkspur

(2) buttercup and larkspur

(3) only buttercup

(4) cotton, coriander, and buttercup

166. Arrange the following layers/regions of the adrenal gland in the correct sequence from outside to inside.

A. Zona fasciculata

B. Adrenal medulla

C. Zona glomerulosa

D. Zona reticularis

(1) D → B → A → C

(2) C → A → D → B

(3) C → B → A → D

(4) A → C → B → D

167. Rop in pBR322 helps in controlling the synthesis of

(1) proteins required for the replication of the plasmid

(2) restriction operator protein

(3) restriction enzyme isolated from *Rhodospirillum* bacteria

(4) a protein involved in the selection of recombinants

168. Which of the following is an oviparous vertebrate that undergoes direct development?

(1) *Neometra* (2) *Neophron*

(3) *Oikopleura* (4) *Orangutan*

169. Given below are two statements.

Statement I : Collenchyma is found either as a homogenous layer or in patches in monocot stem.

Statement II : Sclerenchyma occurs in layers below the epidermis in most of dicotyledonous plants.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

170. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : O_2 can bind with haemoglobin in a reversible manner to form oxyhaemoglobin.

Reason (R) : Each haemoglobin molecule can carry a maximum of four molecules of O_2.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

171. The inner lining of the squamous endothelium of arteries is called

(1) tunica intima

(2) tunica media

(3) tunica externa

(4) elastic lamina

172. The semiconservative mode of DNA replication was first experimentally proved by

(1) Taylor *et al.*

(2) Watson and Crick

(3) Meselson and Stahl

(4) Jacob and Monod

173. The corpus luteum mainly secretes

(1) prolactin

(2) testosterone

(3) glucocorticoids

(4) progesterone

174. The number of chromosomes present in each cell of a dyad after meiosis I is

(1) half that of G_2 phase

(2) same as that of G_1 phase

(3) double that of prophase I

(4) $1/4^{th}$ that of G_2 phase

175. Match List I with List II.

List I (Type of joint)		**List II (Found in)**
A. Saddle joint	I.	Between the carpals
B. Pivot joint	II.	Between carpal and metacarpal of thumb
C. Gliding joint	III.	Between femur and tibia
D. Hinge joint	IV.	Between atlas and axis

Choose the correct answer from the options given below.

(1) A-II B-IV C-I D-III

(2) A-I B-IV C-III D-II

(3) A-II B-IV C-III D-I

(4) A-III B-I C-II D-IV

176. How many of the following are true with respect to EcoRI?

A. It was first discovered, isolated. and characterized as a restriction enzyme among all such type of enzymes isolated from different bacterial cells.

B. Its target site for action is 6 bp of length with four A-T pairs and two G-C pairs.

C. It cleaves the strands of DNA exactly at the middle of the palindrome sequence.

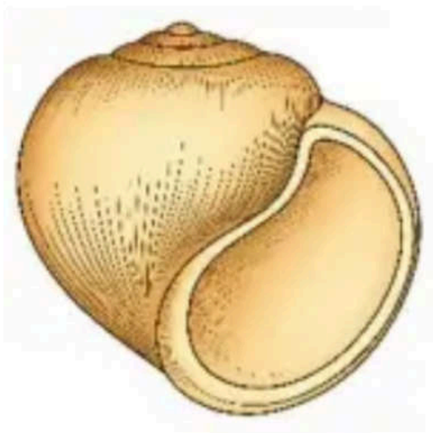
D. It forms the DNA fragments with sticky ends.

E. It breaks the bond between A and A of GAATTC sequence.

(1) 4 (2) 3

(3) 2 (4) 1

177. Observe the figure given below and select the correct statement from the following in relation to the animal.



(1) Body is segmented with a distinct head, muscular foot, and visceral hump.

(2) The circulatory system is of a closed type.

(3) Mantle covers visceral hump.

(4) Antennae, eyes, and parapodia are present.

178. In light reaction, plastoquinone facilitates the transfer of electrons from

(1) PS II to Cyt b_6f complex

(2) Cyt b_6f complex to PSI

(3) PSI to $NADP^+$

(4) PSI to ATP synthase

179. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : The genetic code is degenerate.

Reason (R) : Some amino acids are coded by more than one codon.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

180. Select the INCORRECT statement from the following.

(1) Euglenoids have a pellicle instead of a cell wall.

(2) Slime moulds do not have membrane-bound cell organelles.

(3) Cyanobacteria have pigments studded on chromatophores.

(4) Viruses remain as inert crystals outside the host cell.

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